Can we prove these triangles are congruent? Now we try to prove this using the congruence of triangle rule which we have already learnt. $\angle OCA = \angle OCB = 90°(OC \perp AB)$ and $OA = OB$ is the radius of the circle. The side OC is common. RHS criterion tells us that ΔAOC and ΔBOC are congruent. From this we can conclude that $AC = BC$. This argument leads to the result as follows.

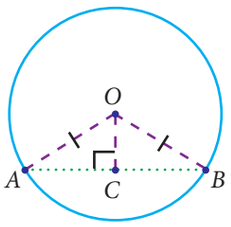


Fig. 4.56

Theorem 7 The perpendicular from the centre of a circle to a chord bisects the chord.

Converse of Theorem 7 The line joining the centre of the circle and the midpoint of a chord is perpendicular to the chord.

Example 4.5 Find the length of a chord which is at a distance of $2\sqrt{11}$ cm from the centre of a circle of radius 12cm.

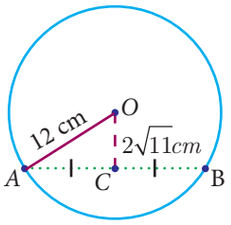


Fig. 4.57

Solution

Let AB be the chord and C be the mid point of AB

Therefore, $OC \perp AB$

Join OA and OC.

OA is the radius

Given $OC = 2\sqrt{11}$cm and OA = 12cm

In a right ΔOAC,

using Pythagoras Theorem, we get,

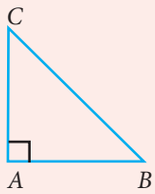
$$AC^2 = OA^2 - OC^2$$
$$= 12^2 - (2\sqrt{11})^2$$
$$= 144 - 44$$
$$= 100\text{cm}$$
$$AC^2 = 100\text{cm}$$
$$AC = 10\text{cm}$$

Therefore, length of the chord $AB = 2AC$

$$= 2 \times 10\text{cm} = 20\text{cm}$$

Note

Pythagoras theorem



One of the most important and well known results in geometry is Pythagoras Theorem. "In a right angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides".

In right $\Delta\ ABC, BC^2 = AB^2 + AC^2$. Application of this theorem is most useful in this unit.

Example 4.6 In the concentric circles, chord AB of the outer circle cuts the inner circle at C and D as shown in the diagram. Prove that, $AB - CD = 2AC$

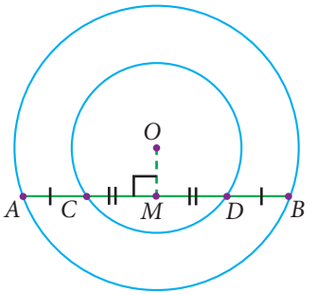


Fig. 4.58

4-Geometry_Term1.indd 163 02/12/2022 21:42:33